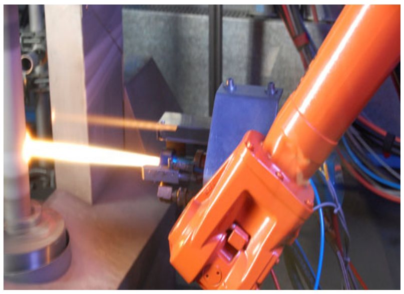
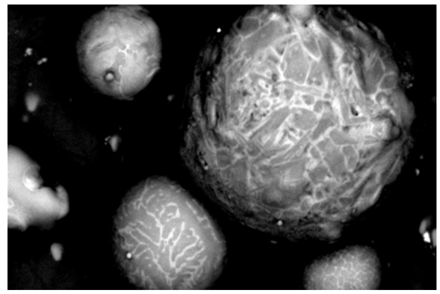
UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Politècnica Superior d'Enginyeria
de Vilanova i la Geltrú

3 EPS Project proposal

Name: Studying and Designing a Laser Treatment Laboratory to Improve HVOF Coatings





Introduction

High-Velocity Oxygen Fuel (HVOF) spraying is a thermal spray technique that produces high-performance protective coatings by accelerating powder particles to supersonic velocities before impacting a substrate. The resulting coatings exhibit low porosity, high density, excellent adhesion, and enhanced mechanical and chemical properties, including outstanding wear, erosion, and corrosion resistance. These characteristics contribute to prolonged component service life and make HVOF one of the most versatile coating methods available, suitable for depositing carbides, metallic alloys, and ceramic materials.

The technique has widespread industrial use in sectors such as aerospace, automotive, marine, and oil and gas, where components are exposed to severe operational conditions. However, despite its many advantages, the HVOF process can introduce microstructural imperfections such as microporosity, oxide inclusions, and partially melted particles, which may reduce the coating's performance. Post-deposition heat treatments are often employed to mitigate these defects, although traditional thermal treatments can undesirably heat the substrate and alter its mechanical properties.

To overcome this limitation, surface-specific treatments, such as laser processing, are increasingly applied to modify only the coating layer while preserving the substrate's integrity. Laser surface treatment offers unique advantages derived from the distinct characteristics of laser radiation, including high beam coherence, precise control of energy input, and flexibility in power density selection. Additionally, this technique is easily automated and can be integrated with thermal spray systems to enhance coating functionality through hybrid or synergistic processing approaches.

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Schedule

First part, up to the midterm defence. In the initial stage of the project, prior to the midterm defence, students will undertake a comprehensive study of HVOF coatings, focusing on their composition, microstructural characteristics, and range of industrial applications. The analysis will include an evaluation of the factors influencing coating performance and durability. Additionally, students will investigate post-deposition treatment techniques designed to enhance coating properties, with particular attention to laser surface treatments as a precise and efficient method for improving the functionality of HVOF-coated components.
Second part, up to the final defence. In the second phase of the project, leading up to the final defence, students will focus on the design and planning of a laboratory dedicated to performing laser treatments on the HVOF coatings under study. This phase involves identifying an appropriate location for the facility and assessing all technical and safety requirements, including electrical connections, network integration and protective measures. All these requirements must be according to the laser power to applied the laser treatments. Students will also design a motorized and programmable X–Y table to enable precise laser processing of the samples. This task includes the selection of suitable high-precision equipment and components, such as a personal computer with dedicated control software, a robust steel framework for securing the laser source, fibber optic connectors, electrical and electronic elements, an aluminium chamber for sample placement and a controlled atmosphere system. Finally, an economic assessment will be conducted to evaluate the overall implementation and operational costs of the proposed laboratory system.

Company

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Contact persons: Joan David Gutiérrez and Silvia Illescas joan.david.gutierrez@upc.edu, silvia.illescas@upc.edu

Project team:

Number of students: 4 to 5, ideally.
Students speciality: Required. At least one student from each of these lines: Industrial engineering, Industrial design. Electrical, electronic, telecommunications, computing… The scope of the project would adapt depending on the final team specialities.